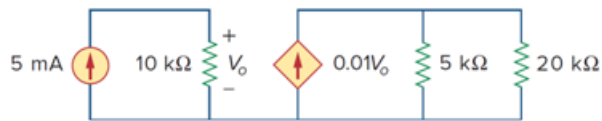


Problem

For the network in Fig. 2.89, find the current, voltage, and power associated with the 20-k Ω resistor.

Figure 2.89:



Step-by-step solution

Step 1 of 5

Draw the circuit diagram.

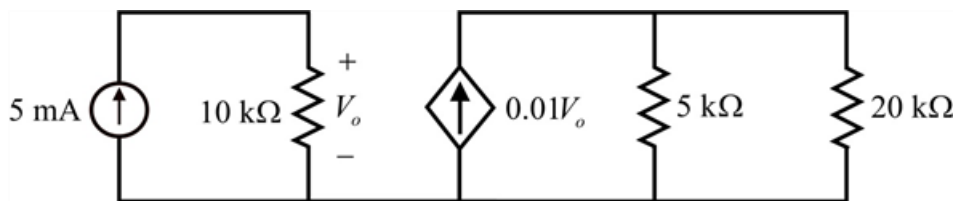


Figure 1

Step 2 of 5

First, calculate the voltage across the 10 k Ω resistor, V_o .

$$V_o = (5 \times 10^{-3})(10 \times 10^3) \\ = 50 \text{ V}$$

Next, find the value of dependent current source, $0.01V_o$.

$$0.01V_o = 0.01(50) \\ = 0.5 \text{ A}$$

Step 3 of 5

Apply current division principle to calculate current through 20 k Ω resistor.

$$I_{20 \text{ k}\Omega} = \left(\frac{5 \times 10^3}{5 \times 10^3 + 20 \times 10^3} \right) (0.01V_o) \\ = \left(\frac{5 \times 10^3}{25 \times 10^3} \right) (0.5) \\ = (0.2)(0.5) \\ = 0.1 \text{ A}$$

Therefore, the current through the 20 k Ω resistor is 0.1 A.

Step 4 of 5

Now, find the voltage across the $20\text{ k}\Omega$ resistor.

$$\begin{aligned}V_{20\text{ k}\Omega} &= (20 \times 10^3)(I_{20\text{ k}\Omega}) \\&= (20 \times 10^3)(0.1) \\&= 2000\text{ V} \\&= 2\text{ kV}\end{aligned}$$

Therefore, the voltage across the $20\text{ k}\Omega$ resistor is $\boxed{2\text{ kV}}$.

Step 5 of 5

Finally, determine the power associated with $20\text{ k}\Omega$ resistor.

$$\begin{aligned}P_{20\text{ k}\Omega} &= (V_{20\text{ k}\Omega})(I_{20\text{ k}\Omega}) \\&= (2000)(0.1) \\&= 200\text{ W}\end{aligned}$$

Therefore, the power associated with $20\text{ k}\Omega$ resistor is $\boxed{200\text{ W}}$.